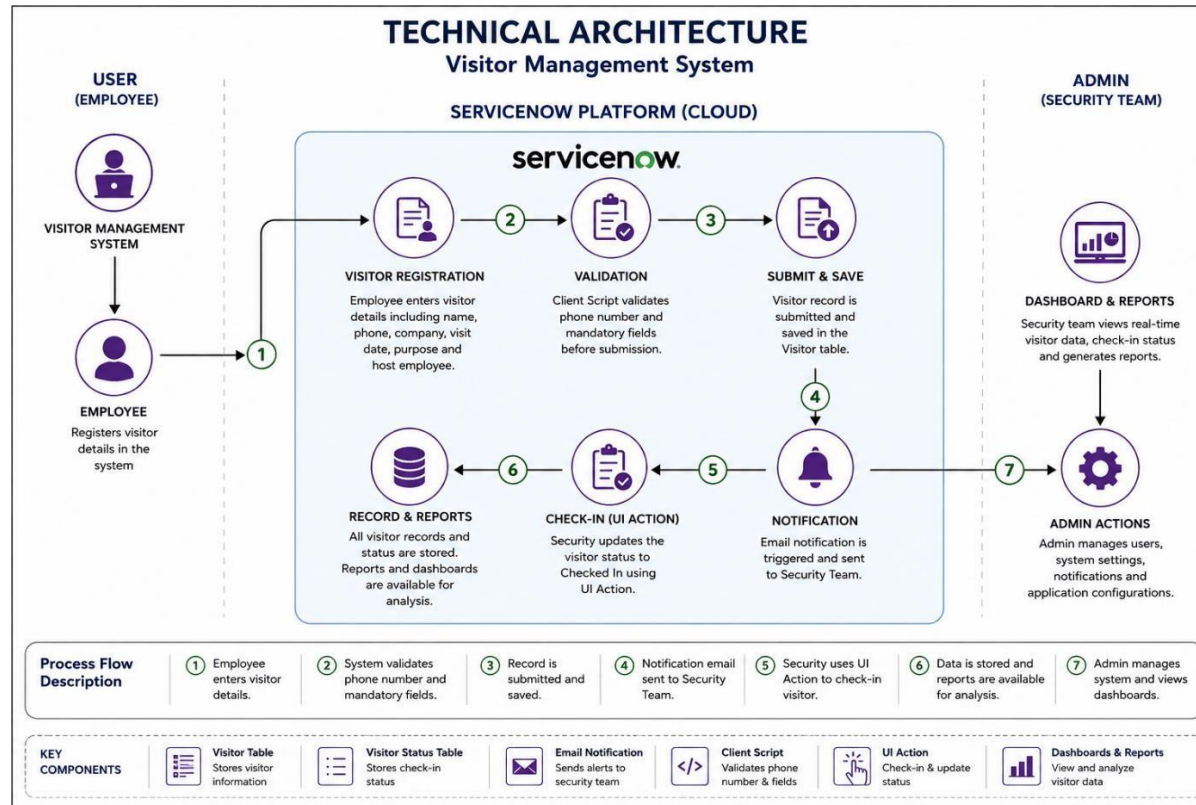


## Project Design Phase-II Technology Stack (Architecture & Stack)

|               |                           |
|---------------|---------------------------|
| Date          | 28-06-2026                |
| Team ID       | 23A51A4211                |
| Project Name  | Visitor Management System |
| Maximum Marks | 4 Marks                   |

### Technical Architecture:



**Table-1 : Components & Technologies:**

| S.No | Component                | Description                                                                                 | Technology / Platform Layer               |
|------|--------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------|
| 1    | User Interface           | Portal interfaces for employee pre-registration forms and front-desk kiosk interactions.    | ServiceNow Service Portal, Portal UI      |
| 2    | Application Logic-1      | Client-side configuration rules verifying valid telephone formats or mandatory fields.      | Catalog Client Scripts (JavaScript)       |
| 3    | Application Logic-2      | Server-side asynchronous retrieval of employee data and validation loops.                   | Script Includes, GlideAjax                |
| 4    | Application Logic-3      | Automated system workflow handling check-in triggers and logging operations.                | Flow Designer, Business Rules             |
| 5    | Database                 | Persistent internal table data structures tracking visitor logs, host relations, and times. | ServiceNow Tables (x_snc_visitor_log)     |
| 6    | Cloud Database           | SaaS cloud-based data tier managed natively by the enterprise platform.                     | ServiceNow Cloud Database                 |
| 7    | File Storage             | Direct document processing storage hosting check-in passes or uploaded ID attachments.      | ServiceNow Attachments (sys_attachment)   |
| 8    | External API Integration | Dynamic verification generation engine or peripheral print utility integrations.            | ServiceNow IntegrationHub / REST Messages |
| 9    | Notification Services    | Transactional engine broadcasting automatic email alerts and SMS confirmations to hosts.    | ServiceNow Notification Engine            |

|    |                          |                                                                                    |                                 |
|----|--------------------------|------------------------------------------------------------------------------------|---------------------------------|
| 10 | Reporting & Intelligence | Analytical widgets providing security teams with real-time daily guest dashboards. | ServiceNow Reports & Dashboards |
| 11 | Infrastructure Host      | Secure, resilient enterprise application hosting and multi-tenant runtime engine.  | ServiceNow SaaS Cloud Platform  |

**Table-2: Application Characteristics:**

| Application Attribute    | Technology Mapping                               | Architecture Implementation Description                                                                                                                    |
|--------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Open-Source Frameworks   | JavaScript, GlideAjax                            | Client-side scripting architectures utilize native JavaScript frameworks to evaluate validation parameters without UI thread freezing.                     |
| Security Implementations | ServiceNow ACLs, User Roles, HTTPS, IAM Controls | Enforces strict row-level Access Control Lists (ACLs) to guarantee only designated security staff and admin contexts can read sensitive data logs.         |
| Scalable Architecture    | ServiceNow SaaS Multi-Tenant Cloud Platform      | Platform scaling handles rapid spikes in lobby foot-traffic or massive increases in logged visitors without requiring infrastructure re-architecting.      |
| High Availability        | ServiceNow Cloud Managed Infrastructure          | Ensures continuous operational uptime for security checkpoints through active redundant hosting nodes and continuous automated platform health monitoring. |
| High Performance         | GlideAjax, Script Includes, Optimized DB Indexes | Asynchronous processing patterns prevent delays by querying database entries over optimized server-side logic scripts instead of locking the browser.      |